

Inference

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Class activity

Work on the class activity, then we will discuss.

Class activity

Question 1: Is there evidence that keeping a pet bird is associated with a greater risk of lung cancer, after accounting for other lifestyle and environmental factors?

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##	Bird	1.36259456	0.41127585	3.3130916	0.00092270
##	SexMale	-0.56127270	0.53116056	-1.0566912	0.29065253
##	SocEcLow	-0.10544761	0.46884614	-0.2249088	0.82205024
##	Age	-0.03975542	0.03548022	-1.1204952	0.26250277
##	SmokeYrs	0.07286848	0.02648741	2.7510612	0.00594025
##	SmokeRate	0.02601689	0.02552400	1.0193110	0.30805533

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Question 2: If an individual does keep a pet bird, is there evidence that smoking further increases their risk of lung cancer, after accounting for other factors?

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```
m_full <- glm(Cancer ~ Bird + Sex + SocEc + Age +  
              SmokeYrs + SmokeRate,  
              data = birdkp, family = binomial)
```

```
m_reduced <- glm(Cancer ~ Bird + Sex + SocEc + Age,  
                 data = birdkp, family = binomial)
```

```
m_reduced$deviance - m_full$deviance
```

```
## [1] 17.66859
```

```
pchisq(m_reduced$deviance - m_full$deviance, df = 2,  
       lower.tail = F)
```

```
## [1] 0.0001456511
```

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Question 3: Consider two individuals. Individual A is male, a non-smoker, of low socioeconomic status, aged 50, who keeps a bird. Individual B is male, of low socioeconomic status, aged 50, who does not keep a bird, and who has smoked for 20 years with an average of 5 cigarettes per day. Estimate the odds ratio of lung cancer for individual A vs. individual B.

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Question 4: Get a 95% confidence interval for the odds ratio.

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```
beta_hat <- coef(m_bird)
V <- vcov(m_bird)
C <- matrix(c(0, 1, 0, 0, 0, -20, -5), nrow = 1)
log_or_est <- C %*% beta_hat
log_or_sd <- sqrt(C %*% V %*% t(C))
exp(c(log_or_est - 1.96*log_or_sd,
      log_or_est + 1.96*log_or_sd))
```

```
## [1] 0.2252356 2.8317351
```